ENGR 202 K-Map of Initialization Fault Report State Machine K-map & Boolean eq-n

Note: V1.01

S0

A \ BC	00	01	11	10
$\bar{A}$ 0	1 (0)	0 (1)	0 (3)	0 (2)
A 1	0 (4)	0 (5)	0 (7)	0 (6)

$$f_{S0} = \overline{(\bar{A} \cdot \bar{B} \cdot \bar{C})}$$

S1

A \ BC	00	01	11	10
$\bar{A}$ 0	0 (0)	1 (1)	0 (3)	0 (2)
1	0 (4)	0 (5)	0 (7)	0 (6)

$$f_{S1} = \overline{(\bar{A} \cdot \bar{B} \cdot C)}$$

S2

A \ BC	00	01	11	10
$\bar{A}$ 0	0 (0)	0 (1)	0 (3)	1 (2)
1	0 (4)	0 (5)	0 (7)	0 (6)

$$f_{S2} = \overline{(\bar{A} \cdot B \cdot \bar{C})}$$

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S3

A \ BC	00	01	11	10
0	0 (0)	0 (1)	1 (3)	0 (2)
1	0 (4)	0 (5)	0 (7)	0 (6)

$$f_{S3} = \overline{(\bar{A} \cdot B \cdot C)}$$

S4

A \ BC	00	01	11	10
0	0 (0)	0 (1)	0 (3)	0 (2)
1	1 (4)	0 (5)	0 (7)	0 (6)

$$f_{S4} = \overline{(A \cdot B \cdot \bar{C})}$$

S5

A \ BC	00	01	11	10
0	0 (0)	0 (1)	0 (3)	0 (2)
1	0 (4)	1 (5)	0 (7)	0 (6)

$$f_{S5} = \overline{(A \cdot \bar{B} \cdot C)}$$

S6

A \ BC	00	01	11	10
0	0 (0)	0 (1)	0 (3)	0 (2)
1	0 (4)	0 (5)	0 (7)	1 (6)

$$f_{S6} = \overline{(A \cdot B \cdot \bar{C})}$$

S7

A \ BC	00	01	11	10
0	0 (0)	0 (1)	0 (3)	0 (2)
1	0 (4)	0 (5)	1 (7)	0 (6)

$$f_{S7} = \overline{(A \cdot B \cdot C)}$$ X

* Realized we only need 7 states, thus the S7 is not being used

* Asynchronous D flip-flops
* A: MSB & C: LSB
* Using 3-input NAND & 1-input NOT gate